

Syllabus
Academic Year 2024-2025

1. General Information	
Course Code	CV4
Course Title	Computer Vision
Degree cycle (level)/ major	Undergraduate, Bachelor of Software Engineering, 6B06102
Year, semester	3-year, 1-semester
Number of credits	5
Language of delivery:	English
Prerequisites	Calculus 1, Calculus 2, Linear Algebra, Probability and Statistics, Introduction to Programming 1, Introduction to Programming 2
Postrequisites	-
Lecturer(s)	Beibit Abdikenov, PhD in Science, Engineering and Technology Assistant Professor, beibit.abdikenov@astanait.edu.kz , Astana IT University, Expo, C1 block, 2nd floor, office C1.1.336
2. Goals, Objectives, and Learning Outcomes of the Course	
1. Course Description	Computer Vision has become ubiquitous in our society, with applications in search, image understanding, apps, mapping, medicine, drones, and self-driving cars. Core to many of these applications are visual recognition tasks such as image classification, localization and detection. Recent developments in neural network (aka “deep learning”) approaches have greatly advanced the performance of these state-of-the-art visual recognition systems. This course is a deep dive into the details of deep learning architectures with a focus on learning end-to-end models for these tasks, particularly image classification. During the 10-week course, students will learn to implement and train their own neural networks and gain a detailed understanding of cutting-edge research in computer vision.
2. Course Goal(s)	Course goal is to provide a solid background in computer vision and deep learning, in preparation either for a job in industry or for more advanced courses at the graduate level.
3. Course Objectives:	Course objectives include: • To understand basics of traditional computer vision and data driven computer vision; • To understand basics of machine learning; • To understand the definition of a range of neural network models; • To understand optimization algorithms for neural network models; • To understand and implement various neural network algorithms in a range of real-world vision applications.

4. Skills & Competences	• Programming. • Analytical. • Problem solving. • Critical thinking. • Communication.
5. Course Learning Outcomes:	Upon completion of the course the student should have the following learning outcomes defined in terms of knowledge, skills and general competence: • explain the basic principles behind neural networks and deep learning; • compare modeling aspects of various neural network architectures; • implement simple neural network algorithms; • apply and evaluate deep learning on real data sets; • provide successful examples how deep learning can be used in different contexts in the society; • read and critically assess papers on artificial neural networks and their applications.
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include assignments and exams. The ungraded activities will be used to monitor progress of students. • Assignment 1 • Midterm • Assignment 2 • Endterm • Final Project
7. Type of teaching	Lectures serve to introduce new concrete examples and solution of them, Furthermore, it develops and provides solving methods based on these examples theoretical and methodological foundations. Practice sessions (seminars) are active sessions to develop student's confidence through new examples and discussions on the problems. Instructor-supervised independent study (ISIS) deals with review and exploration in greater depth of the course material. Student's independent study (SIS): Self-study time including the time required to prepare for and complete all course assessments. Teaching methods: Formal lecture: the lecturer delivers a well-organized, tightly constructed, and highly polished presentation. Lecture-demonstration: The content involves a demonstration of a process or activity. The lecture typically proceeds in chronological order, with the demonstration presented in a sequence of events that the lecturer highlights and explains. Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. Multimedia lecture: Instructors use audio-visual software packages such as PowerPoint or pdf to highlight key points of text.

8. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Lecture presentations. <u>Supplementary literature:</u> 1. Deep Learning with Python, Francois Chollet. 2. Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville. 3. Computer Vision: Algorithms and Applications, Richard Szeliski. 4. CS231n: Deep Learning for Computer Vision.
9. Resources	Online journals, article, papers, books, and internet resources.
10. Keywords	Computer vision, deep learning, convolutional neural networks.
11. Course policy	Course and university policies include: Attendance: Students are expected to attend all scheduled class sessions with all required reading and supplementary materials. Readings are to be completed prior to class. The student won't obtain additional points for course attendance, but the attendance is important to pass the course. In case the student is not able to attend the classes for some reasons, he/she must inform the dean's office in advance and the student itself is responsible for learning all materials, which were given during unattended lessons. In case if the student did not attend more than 30% of the classes without any reasonable excuses, the teacher has a right to mark him as "not graded", and the student wouldn't be admitted to the exam. In other words, students must participate in at least 70% of all online/offline class time, otherwise he/she fails the course. Office hours: 1 hour per week. Every Wednesday from 14:00 to 15:00. Please, send email to arrange a one-to-one meeting to discuss the class using both offline and online. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective. - Contributes to moving the discussion and analysis forward.

- Builds on other comments.

Online class work: The duration of each lecture and practical lesson is 50 minutes.

Offline class work: The duration of each lecture and practical lesson is 50 minutes.

Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student would be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Midterm, Endterm, Final Exams: The exam for the course will be held in different formats: multiple choice exam, handwritten exam, code-based exam, project or combination of those. More details will be provided at the time of exam weeks.

1st and 2nd Attestations grades. In case a student's grade from an attestation is less than 25%, student fails the course.

In the event of some extraordinary event, students should notify the teacher and request an extension of the deadline. If approved, a new date will be given to the student depending upon the circumstances.

Laptops and mobile devices can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will

	be considered a breach of discipline and appropriate action will be initiated by the teacher. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams. 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation. 3. Submitting or using falsified data. 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is only by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, Final Project, etc.*

3. Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Topic and Content	Lectures	Practical lessons	References and Page Numbers	IWS (CPCII) *Tasks and dates
1	1.1. Introduction to Computer Vision. 1.2. Image processing, image formation and geometric vision. 1.3. Introduction to Python.	3	2		
2	2.1. Introduction to Machine Learning. 2.2. Introduction to Pandas and Numpy.	3	2		
3	3.1. Image Classification with Linear Classifiers.	3	2		

	3.2. Regularization and Optimization.				
4	4.1. Introduction to Artificial Neural Networks. 4.2. Backpropagation.	3	2		
5	5.1. Image Classification with Convolutional Neural Networks. 5.2. Convolutional Neural Networks Architectures.	3	2		
Midterm exam					
6	6. Training Neural Networks.	3	2		
7	7.1. Recurrent Neural Networks. 7.2. Attention and Transformers.	3	2		
8	8. Object detection and image segmentation.	3	2		
9	9. Self-supervised learning.	3	2		
10	10. Generative Models.	3	2		
Endterm exam					
	Total hours: 50	30	20		
Final exam					

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	a. Assignment 1 b. Midterm	40 60	100
2 nd attestation	a. Assignment 2 b. Endterm	40 60	100
Final exam	Project	100	100
Total	$0,3 * 1^{st} \text{ Att} + 0,3 * 2^{nd} \text{ Att} + 0,4 * \text{Final Project}$		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

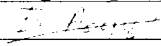
Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught

0-24	No significant assessable material, absent or assessment missing a must pass component
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5. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Beibit Abdikenov	Assistant Professor	26/08/24	

Sergaziyev Muslim
Chair of DCDS

